Replicating a Data-Driven Screening Framework for Secondary and Natural Cementitious Precursors: Code Execution, Modified Training Protocol and Reproducibility Assessment

Original study: Mahjoubi et al. (2025), doi:10.1038/s43246-025-00820-4

Original code: https://github.com/olivettigroup/Cementitious-reactivity-prediction-and-discovery

Replication repository: https://github.com/Shotabdychy/cementitious-reactivity-replication

Abstract

This report replicates the computational workflow of Mahjoubi et al. (2025), who predict the reactivity of cementitious precursors (R^3 heat release, $Ca(OH)_2$ consumption and bound water) with a multi-headed neural network and use it to screen literature-extracted materials and about 10^6 natural rock samples. The released notebooks 9-11 were executed as published and again under a modified training protocol (whole-validation-set early stopping, training-only preprocessing, cross-validated tuning, 11-seed ensembles). Because the authors' input tables and trained weights are not released, inputs were reconstructed from the supplementary datasets, and the language-model extraction step was reproduced on a small sample with a substitute model. As released, the code produced unstable models ($Ca(OH)_2$ test $R^2 = 0.087 \pm 0.130$ over 11 seeds). Under the modified protocol the ensemble reached $R^2 = 0.947$, 0.764 and 0.830 for heat release, $Ca(OH)_2$ consumption and bound water (paper: 0.93, 0.85, 0.89), the latter two resting on only 38 and 56 test labels. Accuracy fell sharply when whole materials were held out ($Ca(OH)_2$ $R^2 = -0.061$). The published rock-screening count was not reproduced, and every comparison that depends on the authors' trained weights is untestable.

1. Introduction

Reducing clinker content requires new supplementary cementitious materials, and screening candidates by measured reactivity is slow. Mahjoubi et al. [1] combine fine-tuned language models that mine oxide compositions from the literature (notebooks 1-8 of their repository [2]) with a multi-task network that predicts three R^3 outcomes from oxide composition, specific gravity, amorphous content, median particle size and test conditions (notebook 9). The network is then applied to 14,429 literature-extracted materials (notebook 10) and to the Gard et al. whole-rock database [3] (notebook 11).

Objective and scope. The aim was to run the released code, obtain its outputs and check the published results within a limited compute budget. Notebooks 9-11 were executed in full. Notebooks 1-8 depend on a private paper corpus and on the authors' fine-tuned endpoints and could not be run; their extraction and classification steps were demonstrated on a small sample of tables (Section 4). Analyses that the paper does not contain (a modified training protocol, held-out-material validation, and an ablation of the rock-screening pipeline) were added independently and are labelled as such. The authors' code [2] and this replication's repository [5] are separate; see Code and Data Availability.

2. Methodology and Computational Implementation

2.1 Source code and reconstructed inputs

The original repository contains 14 notebooks. Five files they load are absent: *R3.csv*, *natural_rocks.csv*, *df_imputed.pkl*, and the trained models *90andupnewones.keras* (notebook 10) and *90andupnewonesch.keras* (notebook 11). Substitutes: (i) *R3.csv* was rebuilt from supplementary Dataset 4 (1,849 of 1,875 rows have a usable material mass; the mix columns, stored in grams, were divided by it); (ii) *df_imputed.pkl* was rebuilt with the notebook's LightGBM imputation of the R^3 features; (iii) *natural_rocks.csv* was rebuilt by merging the Gard et al. tables, keeping all 1,022,092

samples; (iv) both models were retrained, so comparisons with values the authors left in saved cells are reported as untestable, not as failures.

2.2 Faithful execution versus modified protocol

Faithful runs copy notebook 9 cell by cell, including its early-stopping monitor (a normalised RMSE for the $Ca(OH)_2$ head averaged over validation batches of 50 rows), its validation split (the last 15 % of the training array) and its preprocessing, which fits the scaler and target standardisation on all rows before splitting. They were run with 11 seeds (42, 1-10). A clean variant keeps the monitor but removes the leakage and draws the validation rows at random. The modified protocol (an independent addition) changes four things at once: early stopping on the masked validation loss over the whole validation set; a random validation split stratified on which target is labelled; scaling, standardisation and imputation fitted on training rows only; and an outer split stratified on target availability. These changes were not evaluated separately.

2.3 Training, validation and evaluation

Two network configurations were used, and Table 1 states which applies to each row. The released architecture has three NaN-masked regression heads on 256-256-16 ReLU layers, dropout 0.1 after the first layer, Adam at 10^{-3}, batch 50, patience 200 and the notebook's loss weights ($Ca(OH)_2$ head ×5); it is used by the faithful and clean runs and by the "notebook hyperparameters" rows. The tuned architecture was selected by a KerasTuner [4] random search of 18 trials with 5-fold cross-validation inside the training rows, scored on a criterion fixed across trials: widths 512-256-32, dropout 0.2, L2 0.001, learning rate 2.33×10^{-3}, batch 128, equal loss weights with the $Ca(OH)_2$ head ×10, patience 70. All results in Sections 3-4 and the figures use the tuned ensemble unless stated otherwise. The test rows were never read during tuning, early stopping or model selection. Each configuration was trained with 11 seeds and the ensemble is the mean of the seed predictions. The fixed test set holds 370 rows with 264, 38 and 56 labels for heat release, $Ca(OH)_2$ consumption and bound water. Because so few rows carry the second and third labels, 95 % confidence intervals were estimated by a 2,000-sample bootstrap over test rows. To estimate performance on unseen materials, the whole protocol was repeated with a split grouped by material name (277 groups).

2.4 Software environment

Python 3.11.9, TensorFlow 2.15.1, pandas 3.0.6, scikit-learn 1.9.1 and LightGBM 4.7.0 on Windows 11 (8 CPU cores, no GPU), pinned in *requirements.txt*. The paper states Python 3.8, which was not used; one faithful run was repeated with older libraries (TensorFlow 2.12.0, pandas 1.5.3; Section 5). Seeds and commands are in the repository README.

3. Model Replication Results

Faithful runs. All 11 faithful runs satisfy a collapse rule defined on validation behaviour only (monitored value non-finite, restored epoch ≤ 3, or normalised RMSE > 1): the monitored value at the restored epoch ranges from 26.7 to 800. Test R^2 for $Ca(OH)_2$ consumption ranged from -0.120 to 0.360 across seeds, and heat release from 0.584 to 0.876.

Modified protocol. The tuned ensemble exceeds the published heat-release R^2 and is below the published $Ca(OH)_2$ and bound-water values. For those two targets the bootstrap intervals ([0.396, 0.905] and [0.669, 0.910]) include the published values, but this shows only that the test sets are too small to separate the two, not that they are equivalent; the paper reports no intervals and does not state its split. The released hyperparameters scored slightly higher on test than the tuned ones (Table 1); the cross-validation-selected configuration was retained, because choosing by test score would be selection on the test set. The improvement over the faithful runs comes from the combined protocol changes (Section 2.2), whose separate effects were not isolated. The clean variant, which keeps the

original monitor, remains unstable (7 of 11 runs collapse; $Ca(OH)_2$ R^2 = 0.567 ± 0.151), which is consistent with the monitor being an important contributor but does not prove it.

Table 1. Test-set accuracy of the reactivity model: published values and reproduced values. Ensembles are the mean of 11 seeds with bootstrap 95 % confidence intervals in brackets; the faithful row is the mean ± standard deviation over 11 single-seed runs. "Released" architecture: 256-256-16; "tuned": 512-256-32 (Section 2.3). RMSE in J/g (heat release) and g/100 g (other targets).

Row	Heat release R^2 / RMSE	$Ca(OH)_2$ consumption R^2 / RMSE	Bound water R^2 / RMSE
Published (Mahjoubi et al.)	0.930 / 28.20	0.850 / 12.17	0.890 / 1.47
Faithful notebook 9 (released arch.), 11 seeds	0.796 ± 0.087 / 59.98	0.087 ± 0.130 / 29.46	0.828 ± 0.033 / 1.91
Modified protocol, released arch., ensemble	0.956 / 29.14	0.776 / 14.78	0.844 / 1.70
Modified protocol, tuned arch., ensemble	0.947 / 32.02	0.764 / 15.16	0.830 / 1.77
95 % CI of the tuned ensemble	[0.933, 0.960] / [25.32, 38.59]	[0.396, 0.905] / [10.35, 19.92]	[0.669, 0.910] / [1.24, 2.24]
Modified protocol, tuned arch., grouped split	0.645 / 88.68	-0.061 / 48.43	0.548 / 4.56
Test labels (row-level / grouped)	264 / 311	38 / 45	56 / 59

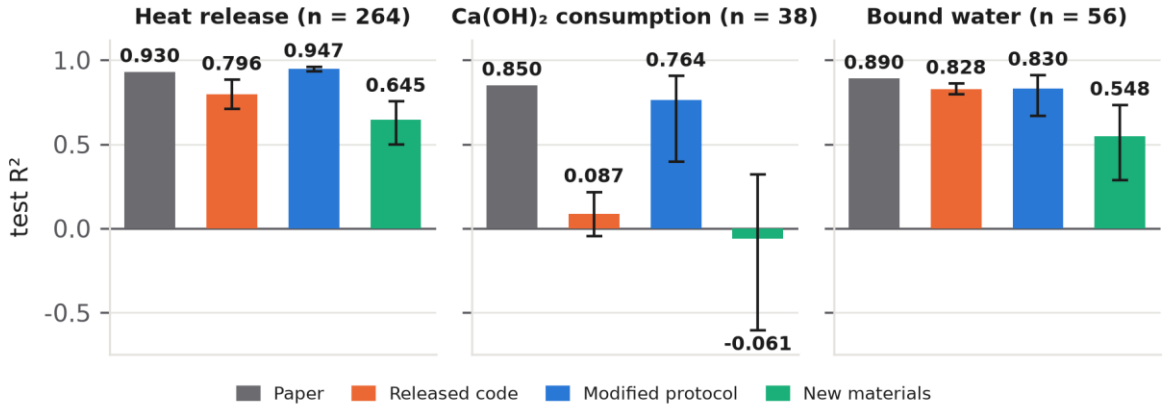


Figure 1. Published and reproduced test R^2 for the three targets (n = labelled test rows). Released code: notebook 9 as published, mean of 11 runs (bar: SD). Modified protocol: tuned 11-seed ensemble (bar: 95 % bootstrap CI). New materials: the same protocol with no material shared between training and test.

4. Literature Extraction and Material Screening

4.1 Extraction demonstration with a substitute model

The authors' fine-tuned GPT-3.5 endpoints are inaccessible, so Claude Opus 5 was used as a substitute extractor with the notebook-4 prompt verbatim. Dataset 2 supplies the table text and a manual ground truth; Dataset 1 supplies the authors' pipeline output, with DOI and caption. Dataset 2 has no DOI, so tables were linked to Dataset 1 by normalised caption (123 of 200 match with a single DOI). Six tables were processed (four composition tables and two non-composition tables); the replies were written before any ground truth was read, and relevance was judged correctly on all six.

Two comparisons, two table sets. Against the *manual ground truth* (Dataset 2), 240 of 241 cells agree exactly, with none missed or invented; this uses rows 191, 107 and 1. Against the *authors' pipeline output* (Dataset 1), 192 of 193 cells agree; this uses rows 191, 105 and 1, because row 107 has no pairable rows (Dataset 1 collapsed its 14 cements into five designations). The two counts therefore overlap in rows 191 and 1 only and should not be compared with each other. The single disagreement in either is the MgO of sample SNM 7: the source table gives 1.57, while both the manual ground truth and Dataset 1 record 1.87. Material typing agreed with the manual top-level type on 36 of 40 unseen Dataset 3 rows (90 %; paper: 97 %), indicative only because Dataset 3 labels are free text (178 distinct values), not the predefined types.

Table 2. Extraction of the four composition tables with the substitute model. "Cells" are (material, oxide) pairs with a value on both sides; the two comparisons use different tables (see text).

Table (Dataset 2 row)	Materials extracted	Exact cells vs. manual ground truth	Pairable with authors' output	Exact cells vs. authors' output
Row 191: Mortar samples	16	111/112	13 of 13	85/86
Row 107: Normal cements	14	84/84	0 of 12	0/0
Row 105: Cement kiln dust	8	not scorable†	7 of 9	62/62
Row 1: Cement and fly ashes	5	45/45	5 of 5	45/45
Total	43	240/241	25 of 39	192/193

† The Dataset 2 ground truth for row 105 describes a different table (a single "Cement" row).

Worked example. Table text and manual values are from Dataset 2 (row 1, which has no DOI); the DOI, category and the authors' output are from the five Dataset 1 rows with the same caption, which cite Tokyay [6]; title and journal are from Crossref. The author confirmed that the table appears in that paper. HL1 is followed through Tables 4-5 using existing records: all nine measured oxides match Dataset 2 and Dataset 1 exactly, and the assigned type (fly ashes) matches Dataset 1.

Table 4. Worked example: HL1, a high-lime fly ash, Tokyay [6]; DOI (from Dataset 1) 10.1016/S0008-8846(99)00160-X. Oxide contents in wt %; - = not reported in the table; italic = imputed.

Source	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	SO_3	TiO_2	MnO	K_2O	Na_2O	LOI
Extracted (Claude Opus 5)	43.3	25.1	4.9	20.3	0.8	2.5	-	-	1.3	0.2	1.19
Manual ground truth (Dataset 2)	43.3	25.1	4.9	20.3	0.8	2.5	-	-	1.3	0.2	1.19
Authors' output (Dataset 1)	43.3	25.1	4.9	20.3	0.8	2.5	-	-	1.3	0.2	1.19
Model input after imputation	43.3	25.1	4.9	20.3	0.8	2.5	*1.18*	*0.17*	1.3	0.2	1.19

Table 5. Assumptions, imputation and predictions for HL1. Parentheses: the same ensemble's prediction for the authors' Dataset 1 row (identical measured inputs; the small difference comes from imputing within a different pool of rows).

Item	Value	Basis
Material type	Fly ashes (Dataset 1: Fly ashes)	assigned from name, caption and descriptor before any label was read
Specific gravity	2.66 g/cm³ (imputed)	not extracted; the table lists 2.25 g/cm³, unused
Amorphous content	54.2 % (imputed)	absent from the table; LightGBM imputation
Fixed inputs	d50 8.88 µm; $Ca(OH)_2$ 3, $CaCO_3$ 0, water 3.6, K_2SO_4 0, KOH 0.151 per g; 50 °C; 10 d	notebook-10 constants, not the table
Predicted heat release, $Ca(OH)_2$ consumption, bound water	417.6 J/g, 76.1 g/100 g, 23.9 g/100 g (411.6, 78.9, 23.4)	11-seed tuned ensemble mean

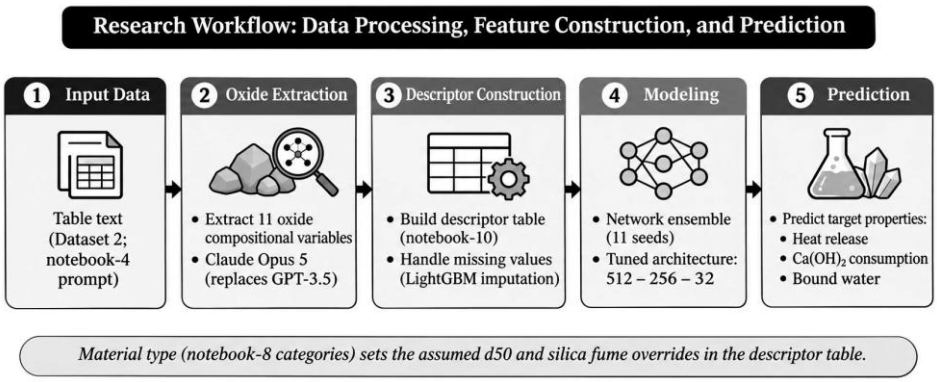


Figure 2. Extraction-to-prediction workflow used in the demonstration; Tables 4-5 follow HL1 through it.

4.2 Literature materials and natural-rock screening

For the 14,429 literature materials the ensemble reproduces the qualitative ordering reported by the paper: metakaolin has the highest median heat release (546 J/g; median $Ca(OH)_2$ consumption 103 g/100 g), fly ashes are pozzolanic (median 70 g/100 g; 87 % above 50 g/100 g), slags and cements show lower $Ca(OH)_2$ consumption (medians 47 and 32 g/100 g), and lime species have the lowest median heat release among the categories discussed (86 J/g). The authors' count of materials above

380 J/g with bound water above 10 g/100 g (964) is untestable without their weights; here it is 4,918, ranging from 2,031 to 6,749 across seeds. 49 predictions exceed the largest training heat release.

The notebook as released, scored with the ensemble over all 1,022,092 samples, calls 141,548 samples reactive (> 200 J/g) against the paper's 50,569, varying from 92,639 to 462,245 between seeds. In the ablation (Table 3) none of the eight combinations of pipeline choices reproduces the paper's figure, and the pozzolanic/hydraulic split depends on a $Ca(OH)_2$ threshold that the released code does not define.

Table 3. Rock samples predicted reactive (> 200 J/g) by the 11-seed ensemble over all 1,022,092 samples, for all eight combinations of three pipeline choices (Dataset 5 in the imputation pool, Fe_2O_3 as total iron, R^3 standard mix instead of the notebook constants). The paper reports 50,569. Counts restricted to samples with SiO_2, Al_2O_3 and CaO, and single-model counts, are in the repository (results/nb11_sound_tuned_ablation.csv).

Change to the notebook	Reactive samples	vs. paper
As released (none)	141,548	+180 %
Dataset 5	74,575	+47 %
Total iron	119,593	+136 %
Dataset 5 + total iron	64,239	+27 %
R^3 standard mix	577,417	+1042 %
Dataset 5 + R^3 mix	586,091	+1059 %
Total iron + R^3 mix	580,477	+1048 %
Dataset 5 + total iron + R^3 mix	567,816	+1023 %

5. Reproducibility Assessment and Discussion

- **Early stopping.** The notebook monitors a normalised RMSE for the $Ca(OH)_2$ head, averaged over validation batches of 50 rows. With 192 labels in 1,849 rows most batches hold none or one label, so the metric is NaN or of order 10^6 and the restored epoch is not meaningful.
- **Splitting and preprocessing.** The stratification key is built on unstandardised targets but applied to standardised ones (six classes), the scaler and target standardisation are fitted on all rows before the split, and validation is the last 15 % of the training array.
- **Provenance.** The saved notebook records about 1,050 executions run out of order, and the model-selection protocol is not specified; Keras Tuner is cited but no tuner call is present.
- **Rock screening.** Notebook 11 scores every row (including 328,466 with no measured oxide), has no pozzolanic/hydraulic rule or rock-name grouping, and does not use Dataset 5; its mix constants differ from the R^3 mix in Dataset 4.
- **Published datasets.** Dataset 2 row 105 has a ground truth from another table, row 191 has unquoted commas in material names, Dataset 3 labels are free text, and the row counts of Datasets 3 and 4 differ from the Supplementary Information.

Published versus reproduced. Heat release is reproduced and the other two targets are partially reproduced under the modified protocol; no faithful run reaches the published $Ca(OH)_2$ value. Counts that depend on the authors' weights (964 materials; 50,569 rocks) cannot be tested. In the rock ablation, adding Dataset 5 to the imputation pool lowers the imputed amorphous content of rocks from 56.4 % to 27.1 % (the measured rocks average 27.1 %) and is the single change that lowers the count most; the R^3 mix multiplies it by about 4. These show the count's sensitivity, not the authors' procedure.

Grouped validation. When no material appears in both training and test, the ensemble gives R^2 = 0.645 (heat release), -0.061 ($Ca(OH)_2$ consumption, 95 % CI [-0.606, 0.320]) and 0.548 (bound water), while training rows are still fitted at R^2 = 0.957 for $Ca(OH)_2$. Dataset 4 has 1,849 rows but only 277 materials, mostly measured at several ages, so a row-level split mainly measures interpolation between ages of materials already seen. The paper does not state its split, so this measures how much the split matters, not an error in a published number.

Limitations. (i) The effects of the individual protocol changes were not isolated. (ii) The $Ca(OH)_2$ and bound-water test sets are small, so single R^2 values are noisy. (iii) The extractor is a substitute and was run interactively, so no timing comparison is made. (iv) Python 3.8 was not used; the library check changed heat-release R^2 from 0.781 to 0.871 but left $Ca(OH)_2$ unchanged (0.128 versus 0.127). (v) A clean-start second run reproduced the headline outputs identically; not every run was repeated.

6. Conclusions and Future Research

The released code runs, but single runs of notebook 9 are not reliable: its early-stopping criterion collapsed in all 11 faithful runs. With a protocol that evaluates validation loss over the whole validation set and avoids preprocessing leakage, the model reaches the published heat-release accuracy and approaches the published $Ca(OH)_2$ and bound-water accuracy within the uncertainty of very small test sets. That accuracy does not carry over to materials absent from training, and the published rock-screening totals could not be reproduced from the released files. The literature-extraction step transferred well to a substitute model on the tables that could be scored.

Potential future research. The following are directions suggested by this work, not experiments that were carried out.

- Screen alternative precursors with validation on held-out material families, reporting grouped alongside random-split accuracy.
- Predict reactivity from oxide composition and descriptors with uncertainty estimates, flagging out-of-range extrapolations.
- Add phase-assemblage descriptors (e.g. QXRD phase fractions) to study how mineralogy and chemistry jointly control hydration performance.
- Rank industrial by-products and natural rocks as supplementary cementitious materials once composition coverage, amorphous content and the pozzolanic/hydraulic criterion are documented.
- Combine the model with experimental characterisation, using model disagreement to choose which materials to test next.

Code and Data Availability

The original implementation is available at the Olivetti Group GitHub repository: https://github.com/olivettigroup/Cementitious-reactivity-prediction-and-discovery (replication used commit 2340ec8). The scripts, modified implementations, computational outputs and supporting documentation developed for this replication study are available in the accompanying replication repository: https://github.com/Shotabdychy/cementitious-reactivity-replication. Certain original datasets, trained model weights and private extraction resources were unavailable; their limitations and the corresponding reconstruction or substitution procedures are documented in this report. The datasets themselves are not redistributed; the README gives their sources.

References

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[3] Gard, M., Hasterok, D. & Halpin, J. A. Global whole-rock geochemical database compilation. *Earth Syst. Sci. Data* **11**, 1553-1566 (2019).

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[6] Tokyay, M. Strength prediction of fly ash concretes by accelerated testing. *Cem. Concr. Res.* **29**, 1737-1741 (1999). doi:10.1016/S0008-8846(99)00160-X